

Pre-Hackathon Consultation — Agenda

Project: Spatial Degradation of Hepatocyte Zonation
Genomics 76553

Team: Roe & Shira

Course: Comp.

One line: use Paper 2's healthy human-liver zonation atlas as a quantitative “ruler” to give Paper 1's spatially-blind disease hepatocytes a pseudo-zonation coordinate, then measure how zonation collapses across MASLD stages (H1), which gene programs collapse (H2), and whether de-zonated cells turn plastic (H3).

Background for the questions. The ruler is a per-cell coordinate = $\text{mean}_z(\text{pericentral genes}) - \text{mean}_z(\text{periportal genes})$, with the gene sets taken from **Paper 2's healthy zonation tables** (genes scored as significantly zoned, assigned to a zone by their center-of-mass). We have four nested sets, all from Paper 2: **full** (~1,600 transcriptome-wide zoned genes), **expanded** (~100 top-ranked), **core** (~13/8 hand-curated), and the **landmark** set — Paper 2's exact 20+20 hepatocyte landmark genes (from its published -LM lists), which Paper 2 itself uses to assign snRNA zonation. The “**validation battery**” = the checks that the coordinate truly measures position: recovery of canonical markers on healthy cells (positive control), within-cell pericentral–periportal anti-correlation, internal coherence of each module, and split-half reproducibility.

1. Methodology & biology (calls that could change direction)

1. **Signature strategy.** We lead with the *full* transcriptome-wide program (the novelty: read position from the whole program, not a few markers), and report the smaller canonical sets as an audit/robustness check; the validation battery decides which leads. Is this appropriately novel without over-reaching?
2. **Classifier ground truth.** We will train the zone classifier on Paper 2's *true Visium-mapped* zonation labels (not a coordinate-derived label), so its per-cell entropy is an *independent* de-zonation signal. Is that the right ground truth, and any pitfalls in the Visium → snRNA label transfer?
3. **Biological prior on the collapse.** Is there a known *directional* pattern (e.g. pericentral injury earlier in MASH) we should test as a specific hypothesis, rather than only a generic monotonic decline?
4. **Is our metric “de-zonation”?** We quantify it per donor as: coordinate spread narrowing, pericentral–periportal anti-correlation weakening toward 0, and classifier entropy rising. Do these capture “loss of zonation” as you would define it?
5. **Plasticity axis (H3).** Are cholangiocyte / ductular-reaction markers (*KRT7*, *KRT19*, *SOX9*) the right readout for hepatocyte plasticity in MASLD, or is there a more established de-differentiation signature?

2. Scope & expectations

1. **What does top-mark work look like?** We aim to deliver *all* of H1–H3 (plus a bonus enrichment) in two days. Given that, what most distinguishes an excellent project here — depth, breadth, rigor of controls, or the write-up / talk?
2. **Small n at the extremes.** Cirrhosis ($n=4$) and end-stage ($n=5$) rest on few donors. Acceptable as a stated limitation with donor-level confidence intervals, or would you handle it differently?